

ASSIGNMENT REPORT-5

MAXIMUM LIKELIHOOD PARAMETER ESTIMATION

Submitted by:

KARTHIKEYAN M [26120004]

MIRTHULA R [261200121]

Under the guidance of:

HEMA ARUNACHALAM MURTHY

For the subject

Pattern Recognition and Machine Learning

(26AI5707)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

SHIV NADAR UNIVERSITY, CHENNAI

OCTOBER 2026

INTRODUCTION

This assignment is all about maximum likelihood parameter estimation (i.e.) we take three 2D Gaussian classes and estimate their parameters, $\hat{\boldsymbol{\mu}}_k$ and $\hat{\boldsymbol{\Sigma}}_k$. Not only do we estimate the parameters, we also split the data into train and test sets and test the estimated parameters, which is estimated from the train set and observe the accuracy, confusion matrix and decision boundary and we repeat this process for different cases of covariance like isotropic, diagonal and full covariance and we also explore shared and different covariances among different classes.

EXPERIMENTAL SETUP

Classes	Three 2-D Gaussian classes
Means	$(-2, -1), (4, 0), (1, 5)$
Samples	10,000 per class
Train/test	8,000 / 2,000 per class
Covariances	Isotropic, diagonal, full; shared and class-specific

PARAMETER ESTIMATION AND CLASSIFICATION

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{N_k} \sum_i \mathbf{x}_{ki}, \quad \hat{\boldsymbol{\Sigma}}_k = \frac{1}{N_k} \sum_i (\mathbf{x}_{ki} - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_{ki} - \hat{\boldsymbol{\mu}}_k)^\top.$$

$$g_k(\mathbf{x}) = -\frac{1}{2}(\mathbf{x} - \hat{\boldsymbol{\mu}}_k)^\top \hat{\boldsymbol{\Sigma}}_k^{-1}(\mathbf{x} - \hat{\boldsymbol{\mu}}_k) - \frac{1}{2} \log |\hat{\boldsymbol{\Sigma}}_k|.$$

Here $\hat{\boldsymbol{\mu}}_k$ is the estimated mean and $\hat{\boldsymbol{\Sigma}}_k$ is the estimated covariance matrix. The function $g_k(\mathbf{x})$ gives us log-density score. Since, the priors are equal here it tells you given the parameters how likely it belongs to class-k.

Taking a closer look into the decision boundaries:

Let $A_k = \hat{\boldsymbol{\Sigma}}_k^{-1}$. Then the squared Mahalanobis distance is represented as

$$(\mathbf{x} - \hat{\boldsymbol{\mu}}_k)^\top A_k (\mathbf{x} - \hat{\boldsymbol{\mu}}_k) = \mathbf{x}^\top A_k \mathbf{x} - 2\hat{\boldsymbol{\mu}}_k^\top A_k \mathbf{x} + \hat{\boldsymbol{\mu}}_k^\top A_k \hat{\boldsymbol{\mu}}_k.$$

Substituting this into $g_k(\mathbf{x})$ gives

$$g_k(\mathbf{x}) = -\frac{1}{2} \mathbf{x}^\top A_k \mathbf{x} + \hat{\boldsymbol{\mu}}_k^\top A_k \mathbf{x} - \frac{1}{2} \hat{\boldsymbol{\mu}}_k^\top A_k \hat{\boldsymbol{\mu}}_k - \frac{1}{2} \log |\hat{\boldsymbol{\Sigma}}_k|.$$

When we compare two classes i and j , the discriminant function is represented like this

$$g_i(\mathbf{x}) - g_j(\mathbf{x}) = -\frac{1}{2} \mathbf{x}^\top (A_i - A_j) \mathbf{x} + (\hat{\boldsymbol{\mu}}_i^\top A_i - \hat{\boldsymbol{\mu}}_j^\top A_j) \mathbf{x} + \text{terms independent of } \mathbf{x}.$$

When different classes share the same covariance, $A_i = A_j$. The quadratic term $\mathbf{x}^\top (A_i - A_j) \mathbf{x}$ gets canceled and leaves us with a linear decision boundary. When different classes have different covariances, that quadratic term remains and produces a nonlinear decision boundary.

ISOTROPIC COVARIANCE: SHARED ACROSS CLASSES

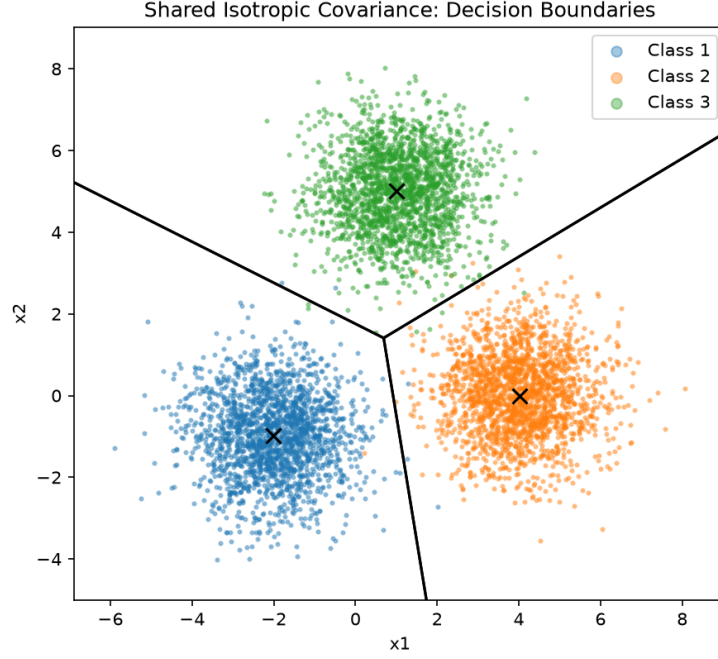


Figure 1: Test data and decision boundary. Accuracy: 99.73%.

Estimated means

Class	$\hat{\mu}_{x_1}$	$\hat{\mu}_{x_2}$
1	-2.01	-0.98
2	4.01	0.00
3	1.00	5.00

Covariance

Class	Generating	Estimated
1	1	1.01
2	1	1.01
3	1	1.01

Confusion matrix

T/P	1	2	3
1	1995	3	2
2	1	1995	4
3	2	4	1994

All these 3 classes share the same covariance matrix and thus, when we define the decision boundary the common quadratic term cancels out as we discussed before and that would leave us with a linear decision boundary. Since all the classes are equally spread and also they have well separated means, the errors are minimal (i.e.) we only have 16 misclassifications out of 6,000 data points from the test set.

ISOTROPIC COVARIANCE: DIFFERENT ACROSS CLASSES

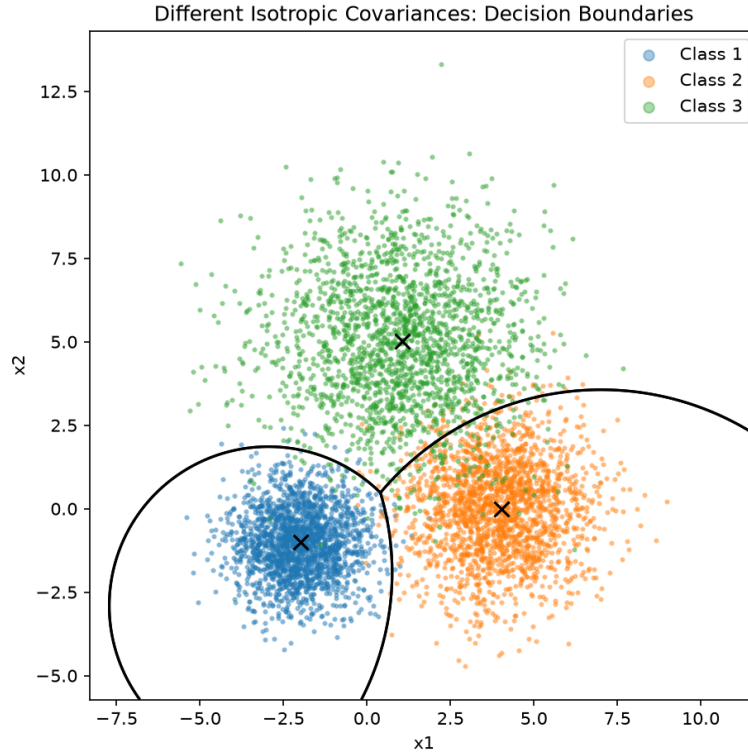


Figure 2: Test data and decision boundary. Accuracy: 96.57%.

Estimated means

Class	$\hat{\mu}_{x_1}$	$\hat{\mu}_{x_2}$
1	-1.98	-0.99
2	4.02	-0.01
3	1.05	5.01

Covariance

Class	Generating	Estimated
1	1	0.98
2	2	2.04
3	4	4.07

Confusion matrix

T/P	1	2	3
1	1981	9	10
2	7	1928	65
3	18	97	1885

Here in this case, the classes have difference covariances and but the features x_1 and x_2 still have equal spread among each of the respective classes. Another thing to note here is, since we have different covariances the quadratic term in the discrimination function stays and thus resulting in a non-linear decision boundary. As you could see the points overlap a bit than before and thus causing some more misclassifications along the way.

DIAGONAL COVARIANCE: SHARED ACROSS CLASSES

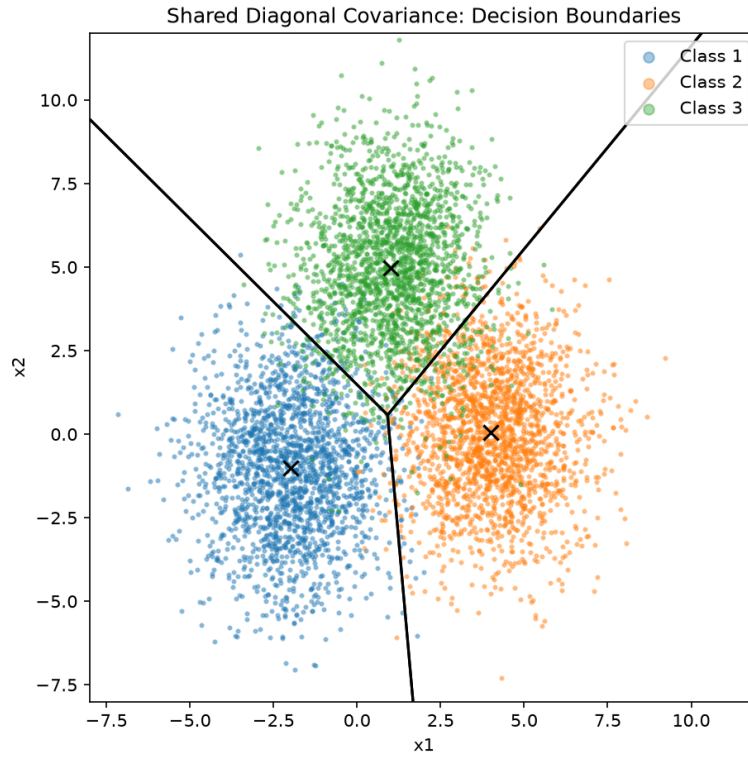


Figure 3: Test data and decision boundary. Accuracy: 93.65%.

Estimated means			Covariance			Confusion matrix			
Class	$\hat{\mu}_{x_1}$	$\hat{\mu}_{x_2}$	Class	Generating	Estimated	T/P	1	2	3
1	-1.99	-1.01	1	(2, 4)	(2.00, 3.97)	1	1908	35	57
2	4.01	0.05	2	(2, 4)	(2.00, 3.97)	2	19	1882	99
3	1.00	4.97	3	(2, 4)	(2.00, 3.97)	3	75	96	1829

Here have work on diagonal covariance (i.e.) the different classes have same covariance matrices, means different classes get the same spread but different variances among x_1 and x_2 directions. Since they have shared covariance matrices the quadratic term gets canceled out and we get a linear decision boundary.

DIAGONAL COVARIANCE: DIFFERENT ACROSS CLASSES

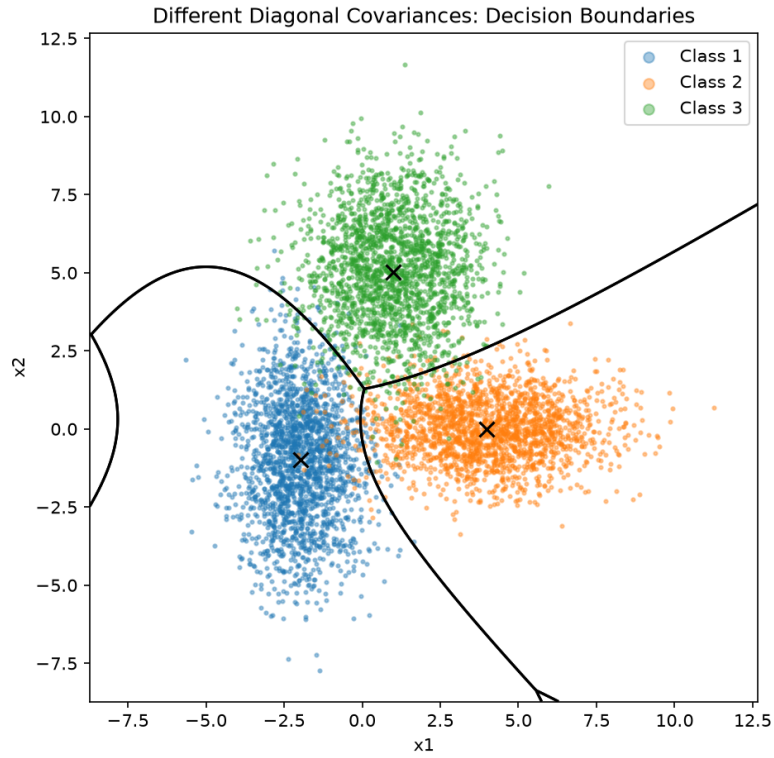


Figure 4: Test data and decision boundary. Accuracy: 96.40%.

Estimated means			Covariance			Confusion matrix			
Class	$\hat{\mu}_{x_1}$	$\hat{\mu}_{x_2}$	Class	Generating	Estimated	T/P	1	2	3
1	-1.98	-0.99	1	(1, 4)	(1.00, 4.09)	1	1943	29	28
2	3.99	-0.01	2	(4, 1)	(4.07, 1.01)	2	44	1927	29
3	1.00	5.02	3	(2, 3)	(2.00, 3.03)	3	40	46	1914

Similarly we have diagonal covariance but different across classes meaning that not only the variance among the features is different, the spread among different classes is also different (i.e.) and as we have already seen the different covariance matrices mean the quadratic term stays in the discrimination function and therefore a non-linear decision boundary.

FULL COVARIANCE: SHARED ACROSS CLASSES

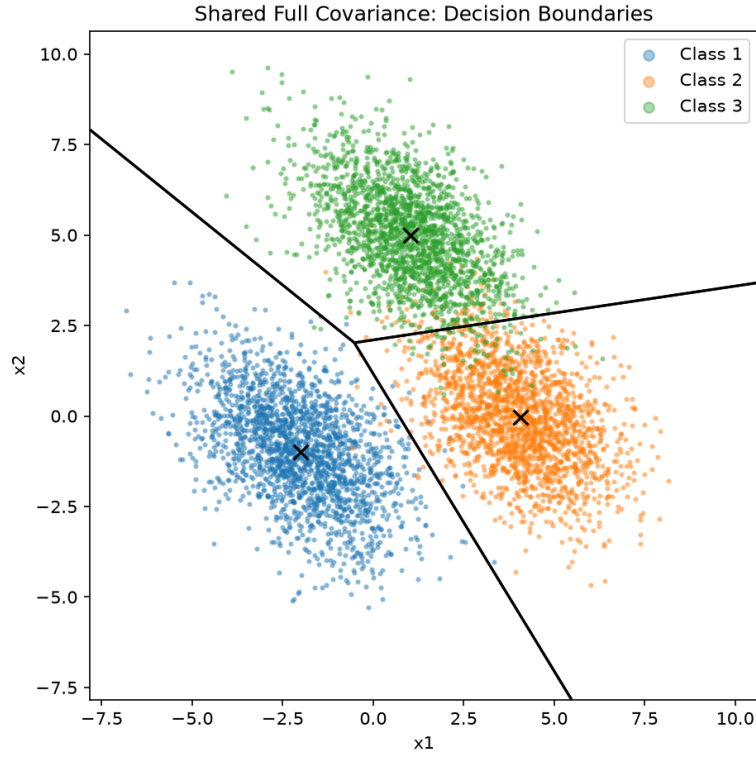


Figure 5: Test data and decision boundary. Accuracy: 97.40%.

Estimated means			Covariance			Confusion matrix			
Class	$\hat{\mu}_{x_1}$	$\hat{\mu}_{x_2}$	Class	Generating	Estimated	T/P	1	2	3
1	-2.01	-0.99	1	(2, -1, 2)	(2.00, -1.00, 2.01)	1	1992	7	1
2	4.06	-0.03	2	(2, -1, 2)	(2.00, -1.00, 2.01)	2	10	1917	73
3	1.03	5.00	3	(2, -1, 2)	(2.00, -1.00, 2.01)	3	2	63	1935

Here, we have a full covariance case means the features x_1 and x_2 have some correlation cause the cloud of data points to tilt, the tilt being positive or negative depends on the sign of the $Cov(x_1, x_2)$ and About the decision boundary, it will be linear because we have shared covariance case, hence, the quadratic terms cancels out.

FULL COVARIANCE: DIFFERENT ACROSS CLASSES

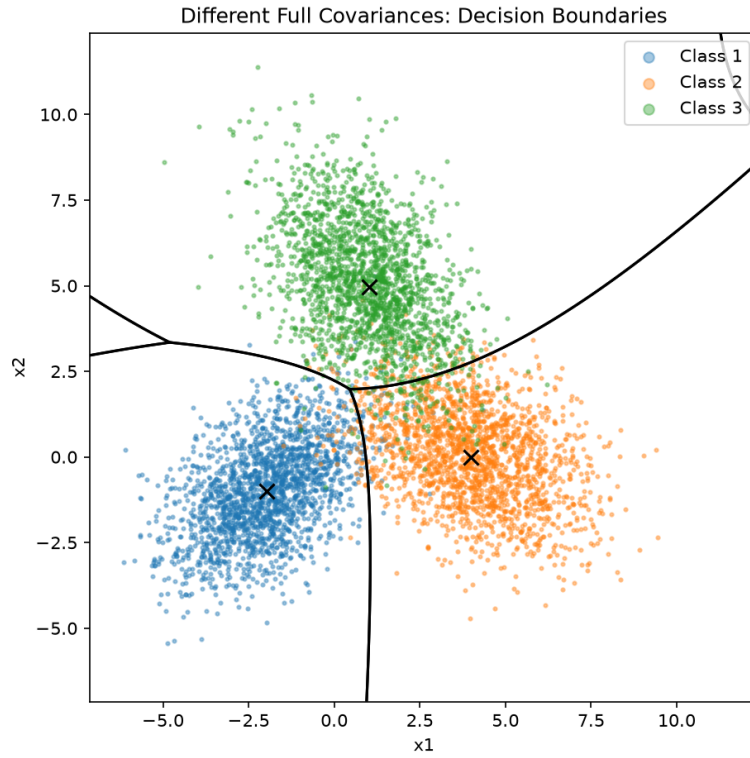


Figure 6: Test data and decision boundary. Accuracy: 95.03%.

Estimated means			Covariance			Confusion matrix			
Class	$\hat{\mu}_{x_1}$	$\hat{\mu}_{x_2}$	Class	Generating	Estimated	T/P	1	2	3
1	-1.98	-1.00	1	(2, 1, 2)	(2.01, 0.99, 1.96)	1	1947	37	16
2	3.99	0.00	2	(3, -1, 2)	(3.00, -1.01, 2.02)	2	47	1859	94
3	1.03	4.97	3	(2, -1, 3)	(1.98, -1.02, 2.98)	3	11	93	1896

Similarly, here we have full covariance matrix and different covariances for different classes and thus we end up with a non-linear decision boundary as the quadratic term survives in the discrimination function.

REAL DATA

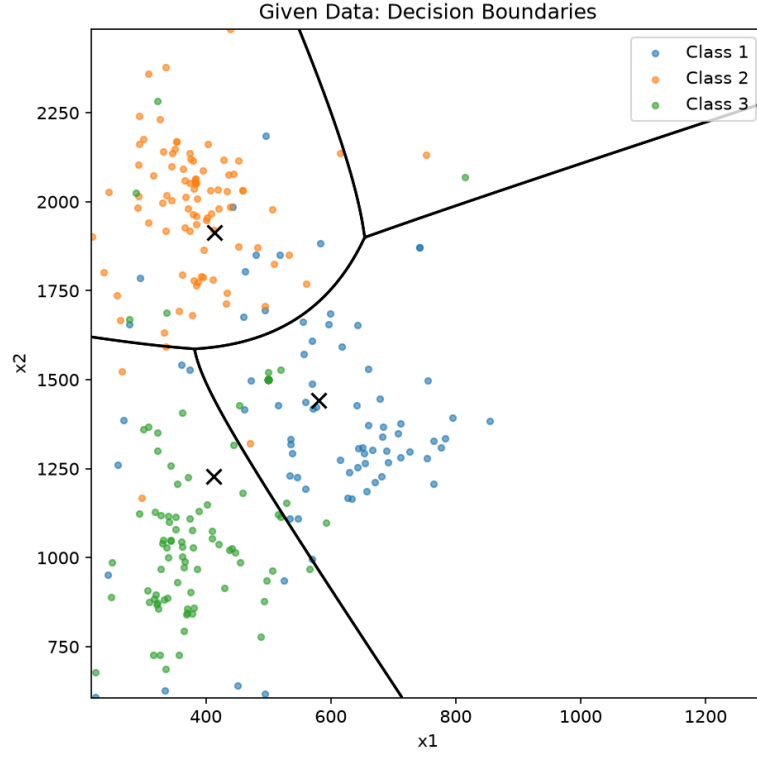


Figure 7: Test data and decision boundary. Accuracy: 80.00%.

Estimated means			Covariance			Confusion matrix			
Class	$\hat{\mu}_{x_1}$	$\hat{\mu}_{x_2}$	Class	Estimated		T/P	1	2	3
1	580.62	1442.99	1	(21843.05, 3347.88, 51710.37)		1	80	10	10
2	413.42	1914.33	2	(13657.83, -9564.81, 47033.19)		2	11	85	4
3	412.50	1227.98	3	(24311.72, 36852.21, 166270.48)		3	21	4	75

Here, we had only very limited number of test points (i.e.) only 100 points for each class and we have a full covariance matrix situation because the off-diagonals are non-zero and The decision boundary is non-linear because different classes they have different covariance matrices and thus the quadratic term survives.

INFERENCE

Covariance	Shared (%)	Different (%)
Isotropic	99.73	96.57
Diagonal	93.65	96.40
Full	97.40	95.03

The main learning curve about these experiments are when you have class overlap they points tends to misclassify here and there and overlap happens when you have different class mean near to each other or even when the mean is a bit away if respective class have different spread for x_1 and x_2 it still might overlap, atleast that's what happened in the diagonal-shared case causing the accuracy to drop to 93.65%.

We were also able to infer when does a decision boundary becomes linear and non-linear by relating it to the discriminant function.